ADNOC GAS OPERATIONS & MARKETING LLC

FLANGE MANAGEMENT PROCEDURE

EPCm CONTRACT NO.: 3000076501
ADNOC GAS PROJECT NO.: 30201-50150-LT-HE

LONG TERM - EXECUTE STAGE

PROJECT 5.0 (P5)

EPCm FOR HABSHAN DBN

A	ISSUED FOR COMPANY REVIEW (ICR)	25-JUL-25	LITHIN Construction Engineer	PRADEESH Principal Construction Engineer	GARY Construction Manager
REV.	DESCRIPTION	DATE	PREPARED	CHECKED	APPROVED
			wood.		

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1.0 INTRODUCTION

1.1 BACKGROUND

ADNOC aims to increase oil production to produce 5.0 MMBOPD by year-end 2027. Accordingly, associated gas production is anticipated to increase. Also, to meet the future oil and gas demand, non-associated gas developments are considered. Hence, the overall gas supply scenarios for major ADNOC Gas (hereafter referred to as COMPANY) assets i.e., Das Facilities, Bab and Habshan Complex (Habshan 0, 1, 2, 4, and 5), Asab and Bu Hasa, is anticipated to change depending on their feed sources, profiles including rates, compositions, and operating conditions. These changes will also subsequently impact the COMPANY downstream facilities, particularly, products handling and transferring facilities i.e., Sales Gas Network, NGL Pipeline Network, Ruwais Fractionation Trains (1, 2, 3, 4 and upcoming new Train 5 part of Bab Cap Gas SELECT Study), Sulphur Handling Facilities, and Condensate systems, etc.

COMPANY has carried out the Long Term DEFINE for the project with M/s Amec International Ltd. (M/s Wood) and subsequently decided to proceed with Long Term EXECUTE phase execution to ensure the readiness of COMPANY facilities to accommodate increased gas supply by December 2027.

Accordingly, COMPANY has awarded EPCm contract to M/s Amec International Ltd. (M/s Wood) to carry out the Long Term scope requirements. M/s Wood shall act as a single entity in the performance of the services and a single point of accountability to ADNOC Gas for the successful completion of Long Term EXECUTE phase.

1.2 LONG TERM EPCM PROJECT OBJECTIVES

The main objective of the Long Term EPCm is to execute the Habshan DBN Project to support ADNOC to achieve oil production target of 5.0 million BOPD so that Associated and Non-Associated Gas can be processed within the overall processing capacity of Habshan. The Long Term EPCm scope includes the following major areas as a minimum:

1. Habshan Complex (H0, H1 Train 1/2, H2, H4, H3 for SS-300 Tie-in)
2. Bab CDS and Bab NGL
3. Habshan 5
4. Pipeline scope activities including below new pipelines
   I. Bab CDS to Habshan Complex HP pipeline
   II. Bab CDS to Habshan Complex MP pipelines (2 no.s)
   III. Habshan 5 to Habshan 2 pipeline

The scope also includes adequacy checks for Ruwais NGL Fractionation Trains, MERAM,

Habshan Sulphur Granulation Plant, Sales gas, NGL, Condensate and interconnecting Pipeline network and Utilities for all impacted AG facilities.

The Long Term EPCM scope includes the following major activities as a minimum:

1. Detailed Engineering and design services
2. Procurement Services
3. Construction Management services
4. Pre-commissioning, Commissioning, Start-up services
5. Final documentation handover and Project close-out

Project also involves major interface activities with other on-going projects. Refer Project Interface management plan (Doc Num: PP-30201_50150_LT_HE-GG-000-50-00-006) for complete list of interface projects.

1.3 SCOPE BATTERY LIMIT

Refer Design Basis (Doc Num: BOD-30201_50150_LT_HE-GG-000-00-00-001) for the scope battery limits, Gas allocation and detailed Design basis.

2.0 PURPOSE OF THIS DOCUMENT

This project procedure defines the process and key stakeholder responsibilities for Quality Assurance and Quality Control (QA / QC) in regard to all flanged joint work completed on pressure containing equipment on P5 Habshan EPCM Project.

In general, all flange joint work performed on any refinery equipment will be subject to QA / QC procedures, with respect to performance and safety before the equipment can be handed over for commissioning.

3.0 ORDER OF DOCUMENT PRECEDENCE

In the event of any conflict between the applicable documents, the order of precedence shall be as follows:

1. UAE Statutory Legislation and Environmental Regulations
2. ADNOC HSE Regulations, HSE Standards and HSE Code of Practice
3. Data Sheets
4. Design Basis
5. Project Specifications and addendum to AGES / DGS (Note-1)
6. COMPANY Standards / Specifications and AGES / DGS (Note-1)
7. International Codes and Standards

Note-1 AGES Addendum/AGES shall be considered wherever AGES addendum/AGES is available. DGS Addendum/DGS shall be considered only in case AGES Addendum/AGES is not available.

However, in case of conflict between the various specifications, standards and the requisition, most stringent requirements shall govern. EPCm CONTRACTOR/VENDOR shall refer any such conflict, and any other matter, which requires further interpretation, to the attention of COMPANY and COMPANY's interpretation shall be final. Resolution of any conflict shall be obtained from COMPANY in writing before proceeding.

Any deviation from COMPANY standard accepted in the existing facility design shall not be considered as basis for this PROJECT design and any deviations shall be submitted to COMPANY for approval before proceeding.

4.0 REFERENCE, CODES AND STANDARDS

4.1 REFERENCE DOCUMENTS

Table 4-1: Reference Documents

Sl. No.	Document Number	Document Title
1.	PP-30201_50150_LT_HE-GG-000-51-00-001	PROJECT QUALITY PLAN
2.	PP-30201_50150_LT_HE-GG-000-51-00-004	CONSTRUCTION QUALITY PLAN
3.	PP-30201_50150_LT_HE-GG-000-51-00-006	QUALITY REQUIREMENTS FOR CONSTRUCTION CONTRACTOR
4.	SP-30201_50150_LT_HE-GG-000-13-00-001	SPECIFICATION FOR BOLT TORQUING TENSIONING PROCEDURE FOR FLANGED CONNECTIONS

4.2 ADNOC STANDARDS

Table 4-2: ADNOC Standards

Sl. No.	Document Number	Document Title
1.	AGES-SP-09-013	Specification for Metallic Pipe, Fittings and Flanges
2.	AGES-SP-09-001	Piping Basis of Design

4.3 INTERNATIONAL CODES AND STANDARDS

Table 4-3: International Codes and Standards

Sl. No.	Document Number	Document Title
1.	ASME B31.3	Process Piping

5.0 DEFINITIONS

5.1 GENERAL DEFINITIONS

For this document, the following general definitions are applicable.

Table 5-1: General Definitions

DEFINITION	DESCRIPTION
COMPANY	ADNOC GAS (AG) for whom the SERVICES are being performed by the EPCm CONTRACTOR.
EPCm CONTRACTOR	Amec International Limited – Abu Dhabi (WOOD) appointed by COMPANY to perform Engineering, Procurement, Construction management, Pre-Commissioning and Commissioning stage for Long Term Solution (under EXECUTE phase).
FACILITIES	Habshan Complex (H0, H1 Train 1/2, H2, H4, H3 for SS-300 Tie-in), Bab CDS, Bab NGL, Habshan 5 and interconnecting pipelines.
LICENSOR	Supplier of packaged units and responsible for the operating and design performance.
MANUFACTURER/ VENDOR/ SUPPLIER	The Party (parties) which manufactures and/or supplies material, equipment, technical documents or drawings and/or services to perform the duties specified by the EPCm CONATRCATOR
PMC	Project Management Consultant appointed by COMPANY for the PROJECT
PMT	Project Management Team
PROJECT	PROJECT 5.0 (P5), Long Term EPCm for HABSHAN DBN
SERVICES	The scope of work to be completed during the EXECUTE stage (EPCm) of the PROJECT.
SHALL	The use of the word “shall” indicate a mandatory requirement

DEFINITION	DESCRIPTION
SHOULD	The use of the word "should" indicate a strong recommendation to comply with the requirements of this document
EPCm CONSTRUCTION CONTRACTOR	Subcontractor appointed by EPCm CONTRACTOR to carry out Construction works at site for respective construction packages.
SUB-CONTRACTOR	The COMPANY approved party, which carries out services subcontracted by EPCm CONTRACTOR.

5.2 TECHNICAL DEFINITIONS

For this document, the following technical definitions are applicable.

Table 5-2:Technical Definitions

DEFINITION	DESCRIPTION
Acid Gas	A gas stream containing predominantly H_2S and/or CO_2 from an Acid Gas Removal Unit
Condensate	A hydrocarbon liquid stream containing Pentane+ compounds
Dehydrated Gas	A gas stream that has been dehydrated to remove water
Dry Gas	A gas stream containing no liquids at flowing conditions, but which may still contain water and hydrocarbons that will fall out if there is a change in flowing conditions.
Feed Gas - Associated Gas	A Gas stream that is produced from an oil well during the extraction of crude oil.
Feed Gas – Non-associated Gas	Hydrocarbon gas originating from gas reservoirs.
GMP 2023	Gas Master Plan 2023
Lean Gas	Lean nature of gas is defined based C2+ in the gas stream. If C2+ is less than 15% then stream is defined as Lean gas
Mixed Feed Gas Stream	Mixed associated and non-associated gas stream

DEFINITION	DESCRIPTION
NGL	A liquid stream with primarily C2, C3, C4 and some C5+ compounds.
Rich Gas	Rich gas is defined based C2+ in the gas stream. If C2+ is more than 15% it is considered rich gas
Sales Gas	Residue gas sent to the Sales gas network.
Sour Gas	A gas stream containing significant amounts of H_2S and/or CO_2.
Sustainable Profile (GMP 2023)	GMP 2023 High Supply Scenario – unconstrained oil production from GMP Guidelines Document from all associated gas fields
Sweet Gas	A gas stream in which the H_2S and CO_2 are not present in significant amounts.
Technical Profile	AON provided profile for technical case for associated gas.
Technical Optimized case	The ADNOC Gas facilities (Das, Asab-0/3, Buhasa, Ruwais, associated pipelines) will be sized to accommodate technical rates from associated gas fields. Habshan facilities will be sized to accommodate technical rate from all associated gas fields (Bab, Buhasa, NEB, Offshore) but not simultaneously. All inlet facilities / inlet compression shall be designed for technical rate.
Unconstrained Production Profile / High Supply Scenario	This shall be based on Upstream's OPEC Unconstrained Oil production case. This scenario corresponds to sustainable projects profiles

5.3 ABBREVIATIONS

ADNOC	Abu Dhabi National Oil Company
AGES	ADNOC Group Engineering Specifications
BOD	Basis of Design

DGS	Design General specification
DEFINE	Front End Engineering Design (FEED)
EPC	Engineering Procurement & Construction
EPCM	Engineering Procurement & Construction Management
FEED	Front End Engineering Design
HSE	Health, Safety and Environment
IFA	Issued For Approval
IFR	Issued For Review
QA/ QC	Quality Assurance/ Quality Control
RFI	Request for Information
ITP	Inspection Test Plan
ITR	Inspection Test Record
MTO	Material Take Off
RMS	Root Mean Square
PED	Pressure Equipment Directive
RTJ	Ring Type Joint
SEP	Standard Engineering Practice

6.0 ROLES AND RESPONSIBILITIES

6.1 COMPANY & EPCM CONTRACTOR

Responsible for:

- Reviewing COMPANY specific QA / QC competence of SUB-CONTRACTOR working with flange joints in pressure containing equipment;
- Witnessing flange makeup or tightening for Special and Critical flanges.

6.2 EPCM CONSTRUCTION CONTRACTOR

6.2.1 Site QA/QC Engineer

The site QA/QC Engineer is responsible for:

- In conjunction with the Piping / Mechanical Superintendent, the QC Engineer is responsible for developing and conducting field audits on SUB-CONTRACTOR activities with flange joints on pressure containing equipment.
- Issuing flange tags and Authorised Fitter stickers to SUB-CONTRACTOR;
- Witnessing flange makeup or tightening for Special and Critical flanges as per Section 7.10.4;
- The QC Engineer is responsible for sign-off of SUB-CONTRACTOR Bolt Torque Test Reports.

6.2.2 Discipline Superintendent

Responsible to the Site EPCM Construction Manager for all Piping / Mechanical field construction activities. Responsible for:

- In conjunction with the QC Engineer, developing and conducting field audits on SUB-CONTRACTOR activities with flange joints on pressure containing equipment;
- Flange inspection for Routine Flanges as per Section 7.8.2 and 7.8.3;
- Ensuring that the methods and timings for the works are technically practical and suitably supported with adequate and appropriate resources (drawings, materials and construction equipment).

6.2.3 Turnover Engineer

Responsible for:

- Inputting and maintaining the flange joint register on the GO Completions Systems Software;
- Ensure the correct ITRs are assigned to the relevant flange joints and signed off in accordance to this procedure and scanning in the Work History Log on completion of joints if required.

6.3 SUB-CONTRACTOR

SUB-CONTRACTOR shall ensure personnel responsible for flange management activities are conversant with this procedure as well as COMPANY / EPCM Contractor procedure.

6.3.1 Authorised Fitter

Responsible for carrying out flange joint work in accordance with the relevant COMPANY & EPCM Contractor QA / QC procedures.

Completes the Work History Log upon completion of their work.

6.3.2 Area Supervisor

Provides initial sign-off of the Work History Log and is responsible for ensuring that the working party have sufficient experience, professional knowledge and training for the work assigned and ensuring that all work is fully completed.

6.3.3 QC Inspector

Provides final sign-off of the Work History Log and is responsible for performing 100% inspections on all flange joint (re)assembled by their company and notifying any nonconformities.

The QC inspector is responsible for completing the Bolt Torque Test Report digitally through the GO Completions Systems Software.

QC inspector is responsible for raising RFI for witness requirements.

7.0 PROCEDURE

7.1 QA/QC Consultation Meeting

Before commencement of any field activities, EPCM CONSTRUCTION CONTRACTOR will hold a QA / QC consultation meeting with the SUB-CONTRACTOR.

SUB-CONTRACTOR personnel that plan to work with flange joints during the Project must have their names, professional experience of mechanical work in refinery, process, or other comparable industries documented. These records must be provided to COMPANY and EPCM CONSTRUCTION CONTRACTOR at least two weeks prior to the project QA / QC consultation meeting.

The QA / QC consultation meeting will provide SUB-CONTRACTOR with pertinent information concerning the current requirements for SUB-CONTRACTOR professional skills, general refinery rules, and project specific procedures that will apply during the Project whilst working with flange joints.

7.2 Authorisation Training

SUB-CONTRACTOR shall have in place a ‘Competency Assurance Program’ to demonstrate suitable training has been provide for personnel involved in Flange and Joint assembly and

inspection works.

7.3 Flanged Joints

The project line list indicates the piping material specification of individual lines. SUB-CONTRACTOR shall use the General Arrangement Drawings or 3D models wherever applicable and MTO to ensure correct information is depicted on Isometrics. Flange categories as per Section 7.8 shall be marked on construction isometrics by SUB-CONTRACTOR (including vessel trim) or within equipment quality packs/ITR (e.g., for manways).

SUB-CONTRACTOR shall also produce a Flange Joint register for input into the GO Completions System Software. Turnover Engineer shall provide a reference Import Table, identifying the required fields to be populated which are required for the Completions System Software. Once the table is imported into the GO Completions System Software, the Turnover Engineer shall assign the relevant ITRs to the flange joints.

SUB-CONTRACTOR shall assemble all joints correctly as per design and engineering requirements, Code and Client requirements with regard to tolerances, alignment, cleanliness, lubrication, work methods, requirements of inspection and provide test records for required joints.

7.4 Tools and Equipment

- Spanner
- Hammer
- Calibrated Torque Wrench
- Hydraulic Wrench
- Tensioning Machine
- Impact Wrenches
- Adjustable Wrench
- Pneumatic Wrenches
- Flange Spreader
- Bolt Tensioning Cylinders

7.5 Bolt Torquing / Tensioning Chart and Sequence

Refer below snaps as mentioned in DGS-1300-160

BOLT TORQUING/TENSIONING CHART

Flange Size NPS (in)	ASME B16.5 Flange Rating																	
	150 lb			300 lb			600 lb			900 lb			1500 lb			2500 lb		
	No. of Bolts	Bolt Dia. (in)	Note No.	No. of Bolts	Bolt Dia. (in)	Note No.	No. of Bolts	Bolt Dia. (in)	Note No.	No. of Bolts	Bolt Dia. (in)	Note No.	No. of Bolts	Bolt Dia. (in)	Note No	No. of Bolts	Bolt Dia. (in)	Note No.
½	4	1/2	1	4	1/2	1	4	1/2	1	4	3:4	1	4	3:4	1	4	3/4	1
¾	4	1/2	1	4	5/8	1	4	5/8	1	4	3/4	1	4	3/4	1	4	3/4	1
1	4	1/2	1	4	5/8	1	4	5/8	1	4	7/8	1	4	7/8	1	4	7/8	1
1 ½	4	1/2	1	4	3/4	1	4	5/8	1	4	1	1	4	1	1	4	1 1/8	1
2	4	5/8	1	8	5/8	1	8	5/8	1	8	7/8	1	8	7/8	1	8	1	1
3	4	5/8	1	8	3/4	1	8	3/4	1	8	7/8	1	8	1 1/8	1	8	1 1/4	1
4	8	5/8	1	8	3/4	1	8	7/8	1	8	1 1/8	1	8	1 1/4	1	8	1 1/2	2
6	8	3/4	1	12	3/4	1	12	1	1	12	1 1/8	1	12	1 3/8	2	8	2	3
8	8	3/4	1	12	7/8	1	12	1 1/8	1	12	1 3/8	1	12	1 5/8	2	12	2	3
10	12	7/8	1	16	1	1	16	1 1/4	1	16	1 3/8	1	12	1 7/8	2	12	2 1/2	3
12	12	7/8	1	16	1 1/8	1	20	1 1/4	1	20	1 3/8	1	16	2	3	12	2 3/4	3
14	12	1	1	20	1 1/8	1	20	1 3/8	1	20	1 1/2	1	16	2 1/4	3			
16	16	1	1	20	1 1/4	1	20	1 1/2	1	20	1 5/8	1	16	2 1/2	3			
18	16	1 1/8	1	24	1 1/4	1	20	1 5/8	1	20	1 7/8	1	16	2 3/4	3			
20	20	1 1/8	1	24	1 1/4	1	24	1 5/8	1	20	2	3	16	3	3			
24	20	1 1/4	1	24	1 1/2	1	24	1 7/8	1	20	2 1/2	3	16	3 1/2	3			
26	24	1 1/4	1	28	1 5/8	1	28	1 7/8	1	20	2 3/4	3						
28	28	1 1/4	1	28	1 5/8	1	28	2	3	20	3	3						
30	28	1 1/2	1	28	1 3/4	1	28	2	3	20	3	3						
32	28	1 1/2	1	28	1 7/8	1	28	2 1/4	3	20	3 1/4	3						
34	32	1 1/2	1	28	1 7/8	1	28	2 1/4	3	20	3 1/2	3						
36	32	1 1/2	1	32	2	3	28	2 1/2	3	20	3 1/2	3						
42	36	1 1/2	1															

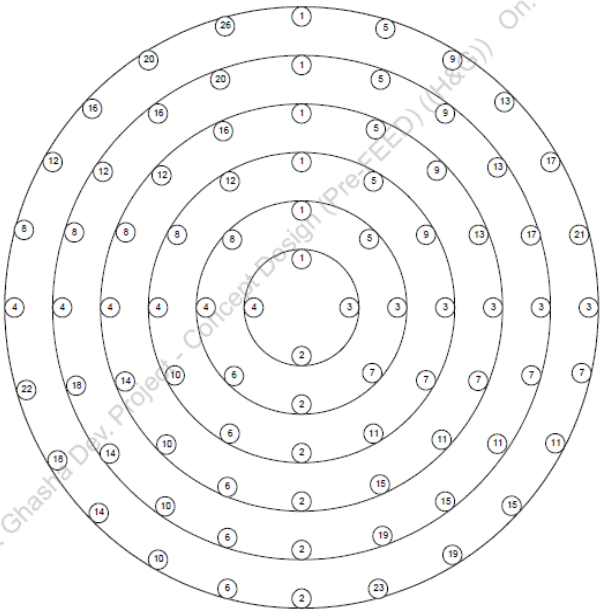
Notes:

1. No Torquing or Tensioning required for these sizes except as per note 4 below.
2. Flange bolting to be tightened to recommended Torque value for these sizes.
3. Flange bolting to be hydraulically tensioned for all services in these sizes. The bolt lengths specified include an additional length of 1 bolt diameter to facilitate the use of bolt tensioning equipment.
4. For all pipe classes in sour service, as a minimum, all the flange bolts shall be tightened to the recommended Torque value:
5. At flanged joints subject to frequent changes in temperature and pressure, and at flanged joints which continue to leak after repeated application of conventional methods.

   Bolt sizes = < 1 3/8" shall be torqued and bolt sizes = > 1 1/2" shall be hydraulically tensioned.

& Chemical Development Project - Concept Design (Rev 0) (Issued On: 5/4/2017

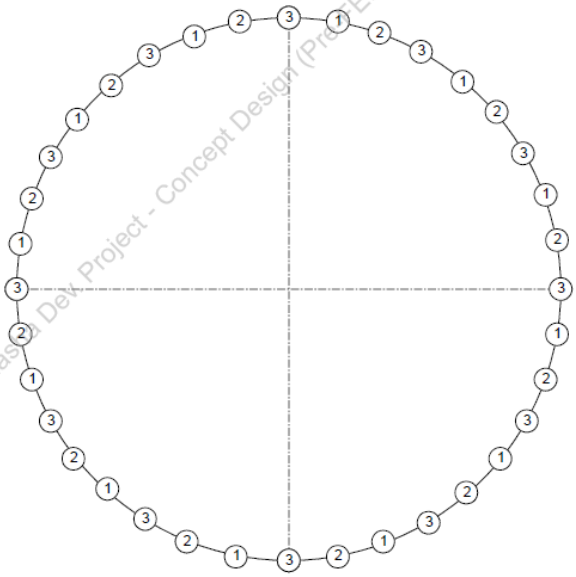
STUD TIGHTENING PATTERN



RECOMMENDED BOLTING-UP SEQUENCE AND NUMBER OF TOOLS TO BE APPLIED

Number of Bolts in Flange	Number of Tensioners
No. of bolts in flange	No. of tensioners to use
64	16
48	12
36	12
24 and less	on a 50% basis

Bolting sequence for 36 bolt flanges
using 12 tensioners



ail & Ghasha Dev. Project - Concept Design (Pre-FEED) ((HE-G)) On: 5/4/2017

7.6 Paint and Pen Markers

Colours for paint and pen markers shall be as follows:

- SUB-CONTRACTOR – White only
- EPCM CONSTRUCTION CONTRACTOR – Yellow and/or Red

Note: use of all paint and pen makers shall be approved by EPCM Contractor QC prior to use.

7.7 Tagging of Flanges

Flange tags shall be provided by EPCM CONSTRUCTION CONTRACTOR to the SUB-CONTRACTOR. SUB-CONTRACTOR shall subsequently distribute these tags to their designated Authorised Fitters. These flange tags are numbered and are unique to each Authorised Fitter. Each SUB-CONTRACTOR must maintain a site record of their Authorised Fitters and their assigned tag number. SUB-CONTRACTOR must only issue flange tags to personnel that have successfully completed the competency training course and the project specific training package described in Section 7.2.

Each flange shall be tagged prior to assembly. Only one flange tag shall be placed per flange/joint. The tags are double-sided and comprise of three distinct sections that are perforated to allow tear off. The tags shall be made of a waterproof material. All details on the tag shall be filled in when applicable and shall be verified during inspection activities. The flange number shall be written on each section in the space provide and shall correspond to the flange number in the Work History Log.

The tag consists of:

Section 1 – is removed once the flange has been assembled and tightened. This section shall be signed by the authorised fitter. Once removed, this shall be returned to the SUB-CONTRACTOR QC department;

Section 2 – is removed once the flange has been torque checked (if required) and approved by SUB-CONTRACTOR QC Inspector shall sign once accepted. Once removed, this shall be returned to the SUB-CONTRACTOR QC department;

Section 3 – shall remain on the flange to show the flange has been inspected, torqued, and approved. Authorised fitter shall sign where applicable as well as the QC Inspector;

If during project activities there is a requirement to perform additional work on a flange that has already been tagged, the tag shall be removed and returned to the EPCM CONSTRUCTION CONTRACTOR QC Engineer as soon as the flange is broken. A new EPCM CONSTRUCTION CONTRACTOR Flange Status Tag shall be placed prior to any work commencing by the Authorised Fitter.

At no time, other than described above, or in consultation with the EPCM CONSTRUCTION CONTRACTOR QC Engineer and the Construction Manager, shall personnel remove flange tags affixed to equipment.

Following the completion of SUB-CONTRACTOR work on the Project, any unused tags assigned to their Authorised Fitters shall be returned to the EPCM Contractor QC Engineer.

7.8 Flange Categorisation

Flanges and Flange assembly shall fall into one of the following categories.

Category	Description
Routine	• PED Group 2 Category SEP, I, II, III; • PED Group 1 Category SEP; • Bolt diameter ≤1"; • And not categorised by EPCM Contractor Construction or Engineering as Special or Critical flange;
Special	• PED Group 1 Category I

	• Reinstatement flanges - which are made after hydro-test and therefore will not be subject to leak test prior to turnover; • Flanges located in special locations: located in inaccessible locations which pose a hazard in the event of a leak regardless of the service fluid; ≥600# in a high-risk location; • Routine flanges with bolt sizes of ≥ 1 1/8"; • Any flanges which have had a known problem in the past (e.g. flange that has leaked during testing and must be remade, flanges where quality observations have been identified and corrected such as minor scratches on face, misalignment etc.); • And not categorised by EPCM Contractor Construction or Engineering as Critical flange;
Critical	• PED Group 1 Category II, III and IV; • All tensioned flanges; • PSV inlet and outlet flanges; • All rotating equipment flange connections with piping (following API 686 guidelines for flange bolt-up); • All RTJ and Grayloc flanges;

7.9 Quality Requirements

SUB-CONTRACTOR shall reference the relevant EPCM CONSTRUCTION CONTRACTOR forms in their ITPs where applicable. SUB-CONTRACTOR may be required to develop and supply additional check sheets and or forms for EPCM CONSTRUCTION CONTRACTOR approval where the available forms are not adequate.

SUB-CONTRACTOR shall ensure their work is signed off in accordance with their approved ITPs where applicable. EPCM CONSTRUCTION CONTRACTOR Discipline Superintendent shall inspect the required Joints and countersign the forms and where required the contractors shall notify the EPCM Contractor and COMPANY QC in accordance with the ITP and this procedure.

Every Flange shall subject to:

- Flange face inspection;

- Lubricant application;
- Alignment (parallelism, offsets etc.) checks;
- Nuts, Stud/Bolt and Washer checks;
- Gasket checks;
- Bolt Load checks

7.9.1 Preliminary Checks

7.9.1.1 Check condition of gasket:

- Visually examine gaskets prior to installation to assure they are free from defects. All damaged gaskets (including loose spiral windings) should be rejected and returned to the warehouse for replacement and recording of stock purposes. Rejected gaskets shall be disposed of appropriately;
- Used gaskets shall not be reused. All gaskets should be replaced with new ones whenever a joint is to be closed;
- Check the gasket colour coding, material and dimensions are in accordance with the applicable job specifications/drawings;
- Clean gasket seating facing using a wire brush (The brush used shall have stainless steel bristles for alloy components);

7.9.1.2 Check condition of stud/bolts, nuts, and washers:

- Check that stud/bolts, nuts and washer material, dimensions including length and type are in accordance with the applicable job specifications/drawings;
- Studs and nuts shall be clean and free of debris;
- Visually examine studs and nuts after cleaning to assure they are free from defects/burrs. All damaged items should be rejected and returned to the warehouse to ensure inventory whereupon they can then be disposed of;
- Nuts should turn freely on the studs for a distance equal to their in-service make-up.
- Where required lubricate the threads and nuts to flange contact surfaces with Molykote 1000;

7.9.1.3 Check condition of flanges:

- Check type, rating, and material are in accordance with the applicable job specifications/drawings;
- Check that flange face surface finish, including facing type, RMS (gramophone finish), alignment is in accordance with the applicable job specifications/drawings;
- Check flange faces for scratches, dirt, scale, remnant gasket and protrusions. Flange faces with coarse tool marks or scratches may form leakage paths and may require re-machining or fitting with an alternative style gasket. Any damaged flange shall be inspected by Wood QC Engineer prior to proceeding with the joint makeup. Flange shall be dry without any lubricant;

- Check flange surface flatness: Use a straight edge or steel rule to check facing flatness. Warped or badly corroded flanges should be replaced. (Wood QC Engineer shall be notified);

7.9.2 Prior to gasket insertion

Check flanges assembly alignment (alignment can be checked using a straight edge or steel rule):

- Check flanges assembly alignment for flanges faces parallelism and axial alignment;
- Check flange assembly gap (sufficient to insert the gasket). Once aligned, visually examine them to ensure an acceptable fit has been obtained;
- Check bolts holes alignment of the mating flange to ensure easy insertion of bolts without any interference with the body of the flange due to flange rotation;
- If necessary, alignment should be achieved by cutting and re-welding the pipe where possible. When aligning flanges, checks shall be made to ensure there are no residual stresses in the joint, particularly in machinery piping;
- Final alignment shall be recorded for reference.

7.9.3 Gasket and bolt insertion

- Visually examine gaskets, prior to installation, to assure they are free from defects. Do not use sealing compounds, grease or other paste or adhesives on the gasket or flange faces. This may cause leaks;
- Visually examine the flange alignment to ensure that an acceptable fit has been obtained. Studs inserted through the bolt holes of the flanges should guarantee the sustained alignment of the flanges without blocking the insertion of the gasket;
- Insert the gasket between the flanges carefully to assure proper placement making certain the gasket is properly cantered in joint;
- Insert all remaining studs and nuts on hand tight. Do not pull up studs in a manner that reaches uneven compression of the gasket;
- Ensure all bolts are installed with at least two threads protrusion through the nuts at both sides. Where bolt tensioner is to be used, threads protrusion shall be at least one bolt diameter on one of two sides;

7.9.4 Bolting Procedure

Bolt procedure as per SP-30201_50150_LT_HE-GG-000-13-00-001.

7.10 Flange Assembly Witnessing

7.10.1 General Requirement

Prior to work on a flange joint commencing, the EPCM CONSTRUCTION CONTRACTOR Area Supervisor must ensure that the working party selected to complete the task have:

- Successfully completed the competent training course;
- Been issued with unique numbered flange tags for each Authorised Fitter;
- Access to complete the Work History Log.

Once the Authorised Fitter has completed the work and affixed the COMPANY Flange Tag onto the flange joint and filled in the relevant sections of the EPCM CONSTRUCTION CONTRACTOR Flange Status Tag, they shall complete all applicable sections of the Work History Log, a pro-forma of which is contained within Attachment 1 for information. Following this, they shall seek appropriate signoff from their Area Supervisor.

The EPCM CONSTRUCTION CONTRACTOR Area Supervisor shall review the work undertaken and only if they deem the work to have been fully completed shall they sign the Work History Log as acknowledgement.

If they deem the work to be incomplete, they should inform the working party as to the outstanding activities.

Once the EPCM CONSTRUCTION CONTRACTOR Area Supervisor has signed the Work History Log to signify the work has been fully completed, they shall seek the appropriate signoff from their QC inspector.

The EPCM CONSTRUCTION CONTRACTOR QC inspector shall complete a 100% inspection on the flange joint that has been assembled/re-assembled. If nonconformities are found, the EPCM CONSTRUCTION CONTRACTOR QC inspector shall record a note in the Work History Log and report the deviation to the Area Superintendent for immediate action.

On notification of a nonconformity, the Area Supervisor shall instruct the working party as to the remedial action to take.

Once the nonconformity has been addressed the EPCM CONSTRUCTION CONTRACTOR Area Supervisor shall make a note on the Flange Working Log of the probable causes for the deviation and subsequently seek the appropriate signoff from their QC inspector.

If no non-conformities are noted during the inspection, they shall sign the Work History Log to signify the flange joint has been approved and ensure that the accurate test result data is recorded on a Bolt Torque Test Report, a pro-forma of which is contained within Attachment 2 for information.

The Work History Log provided in Attachment 1 is an example only and the format is subject to change. The form as a minimum will require the following information to be filled in by EPCM CONSTRUCTION CONTRACTOR:

- Flange Number
- Flange Diameter
- Flange Rating
- Flange Type
- Gasket Type
- Tightening Methodology
- Completed by

- Approved by
- QA / QC Approved by
- EPCM CONTRACTOR Approved by (if required)

The Turnover Engineer may provide pdf copies of the forms to the Contractor which include a barcode to allow upload into the GO Completions System Software.

7.10.2 Routine Flanges

Routine Flanges shall follow the requirements of Section 7.10.1 as well as the requirements below.

Once the EPCM CONSTRUCTION CONTRACTOR's QC Inspector has approved the flange assembly, EPCM CONSTRUCTION CONTRACTOR shall raise an RFI (stating the Flange Category as per Section 7.8) for the EPCM CONTRACTOR Discipline Superintendent to inspect the joint. EPCM CONTRACTOR Discipline Superintendent shall inspect the first 10% of all Routine Flanges made by each Authorised Fitter, thereafter at the discretion of the Superintendent.

When the EPCM CONTRACTOR Discipline Superintendent has completed his checks and has approved the joint, Contractors QC Inspector shall physically mark the nuts with a line across the stud/bolt end, nut and flange hub using an indelible white paint marker, indicating the joint is tight and has been checked.

EPCM CONTRACTOR Discipline Superintendent shall sign the Work History Log.

EPCM CONTRACTOR QC Engineer shall carry out random checks and shall report any issues found to the EPCM CONTRACTOR Discipline Superintendent. If further action is required, this shall be at the discretion of the EPCM CONTRACTOR Discipline Superintendent, EPCM CONTRACTOR QC Engineer and Construction Manager.

EPCM CONTRACTOR Discipline Superintendent shall ensure the work has been carried out in accordance with this procedure and the appropriate documentation and records completed.

7.10.3 Routine Flanges with Calibrated Equipment

Witnessing of routine flange assembly using calibrated wrenches shall be carried out as per Section 7.10.2.

7.10.4 Special and Critical Flanges

Special and Critical Flanges shall have additional inputs by both EPCM CONTRACTOR and COMPANY QC representatives.

For the following activities, EPCM CONSTRUCTION CONTRACTOR shall issue an RFI (stating the Flange Category as per Section 7.8) to EPCM CONTRACTOR QC and/or COMPANY QC for witness. For Critical flanges, these points shall be WITNESS points in the ITP for COMPANY QC:

1. Flange face inspection, fit-up, and the 1st pass (30%) of Torquing (Critical);
2. Final click inspection and sign-off upon completion (Special and Critical);

COMPANY QC have the right to waive inspection requirements under their discretion, but EPCM CONTRACTOR QC shall still be required for the witness points.

If no non-conformities are noted during the inspection, EPCM CONTRACTOR and COMPANY shall sign the Flange Make Up Checklist as per Attachment 3 through the GO Completions System Software. EPCM CONTRACTOR shall physically mark the nuts with a line across the stud/bolt end, nut and flange hub using an indelible yellow or red paint marker, indicating the joint has been witnessed. Subsequently, If the joint is broken, the paint markings shall be removed, and Wood shall be notified and the Flange Make Up Checklist cancelled/superseded.

EPCM CONTRACTOR Discipline Superintendent shall ensure that all requirements have been met prior to an inspection by Preem QC. This shall include:

- Sufficient notice is given to COMPANY QC Inspector;
- Checking the appropriate tools are in place and operable;
- Documentation is appropriately signed off and made ready for any subsequent sign offs by COMPANY QC;
- All personnel required to operate equipment or tension the joint are at the job site;
- Where possible, the tightening pattern numbering to be used is marked on the flange;
- Suitable safe access is available;
- Requirements of Section 7.9 have been met;

8.0 RECORDS

Copies of the EPCM CONSTRUCTION CONTRACTOR's Work History Logs shall be issued to EPCM CONTRACTOR and COMPANYQC upon request. EPCM CONSTRUCTION CONTRACTOR shall also provide copies of the completed Work History Logs to the Turnover Engineer when requested such that they can be scanned into the GO Completions System Software and allocated to the relevant joint for completion tracking.

Copies of the EPCM CONSTRUCTION CONTRACTOR's Bolt Torque Test Reports and Flange Make Up Checklist shall be completed through the GO Completions System Software for subsequent sign-off.

EPCM CONSTRUCTION CONTRACTOR QC inspectors shall report any non-conformities they discover to the EPCM CONTRACTOR QC personnel on a daily basis.

Each EPCM CONSTRUCTION CONTRACTOR must maintain a site record of which of their personnel have completed the competent training module together with a copy of their training certificate and assigned tag number.

APPENDIX-01 FLANGE JOINT WORK HISTORY LOG

wood.

FLANGE JOINT WORK HISTORY LOG

ADNOC

Company:	Work Order:	Location:	Flange Joint ID: *(Blind list etc.)*

Flange No:	Flange Dia:	Flange Rating	Flange Type	Gasket Type: *(SPW / RTJ)*	Tightening Method: *(Wrench / Torque / Bolt Tightening)*	Completed By: *(Name & Tag No)*	Supervisor Approved: (Supervisor Name)	QA / QC Approved: (QA Inspector Name)	Wood Approved (if required)	Notes of Any Deviations:

APPENDIX-02 BOLT TORQUE TEST REPORT

wood.

Project Title
Project Description
BOLT TORQUE TEST REPORT

Page:

SYSTEM NO: | EPCM CONTRACTOR CONTRACT No:

TEST PACK NO: | MAIN P&ID:

1.0 CALIBRATION DATA

Manufacturer: | Calibration Certificate Available: Yes / No

Torque Wrench Range: | Certificate Number:

2.0 RESULTS

Joint No	Bolt Size	Qty	Torque Readings (Nm)	Joint No	Bolt Size	Qty	Torque Readings (Nm)

Remarks' / No Remarks' (*Delete as applicable)

COMPLETED BY	EPCM CONSTRUCTION CONTRACTOR	EPCM CONTRACTOR	Owner
COMPANY			
SIGNATURE			
NAME			
DATE			

APPENDIX-03 FLANGE MAKE-UP CHECKLIST

P5 HABSHAN EPCM PROJECT

FLANGE MAKE-UP CHECKLIST

Page:

SYSTEM NO:			WOOD CONTRACT NO:	1-PD-XXXXX
TESTPACK NO:			MAIN P&ID:	
Critical Flange	YES	NO	CONTRACTOR FLANGE CHECK	WOOD FLANGE CHECK
Non-Critical Flange	YES	NO	FLANGE / GASKET / BOLTING / TORQUE	WOOD MECHANICAL SUPERVISOR

FLANGE JOINT REFERENCE NO ②	IS SERRATION SURFACE IN GOOD CONDITION?①	IS GASKET SURFACE CLEAN? ①	IS FLANGE ALIGNMENT WITHIN TOLERABLE LIMITS?①	IS THE GASKET NEW? ①	IS IT CORRECT TYPE OF GASKET①	IS IT CORRECT SIZE OF GASKET①	IS IT CORRECT RATING OF GASKET①	IS THE BOLT PROTRUSION WITHIN REQUIREMENTS?①	ARE THE THREADS LUBRICATED?①	ARE THE BOLTS OF CORRECT SIZE?①	ARE THE BOLTS/NUTS CORRECT TYPE OF MATERIAL?①	ARE THE BOLTS TIGHTENED IN CORRECT SEQUENCE? ①	IS CONTROLLED TORQUE TIGHTENING REQUIRED? ③	STATE INITIAL TORQUE/BOLT TENSION VALUE ③	STATE INTERMEDIATE TORQUE/BOLT TENSION VALUE ③	STATE FINAL TORQUE/BOLT TENSION VALUE ③	IS HYDRAULIC TORQUE REQUIRED AND IS THE REPORT AVAILABLE?③	ARE FLANGE ALIGNMENT/GAP WITHIN REQUIREMENTS?①	IS THE GASKET OF CORRECT SIZE/RATING/TYPE OF MATERIAL?①	IS THE BOLT PROTRUSION WITHIN REQUIREMENTS?①	ARE THE BOLTS/NUTS OF CORRECT SIZE/LENGTH/TYPE OF MATERIAL/GRADE? ①	ARE THE BOLTS/NUTS ASSEMBLY CHECKED/THREADS LUBRICATED?①	FLANGE TORQUE CHECK ③	WOOD INITIAL

COMMENTS

Notes:

① To fill in Y (YES) or N (No) or N/A

② Critical Joint No shall be defined by Wood Engineering with reference drawing

③ See Torque Requirement

COMPLETED BY	Contractor	WOOD	PREEM (if applicable)
COMPANY			
SIGNATURE			
NAME			
DATE			